

Kritarth Dandapat

PhD Student / Research Assistant, Michigan State University

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Professional Summary

Incoming PhD student in Computer Science at Michigan State University, focused on spatiotemporal machine learning, deep learning based weather forecasting, and adversarial robustness. I moved into research early in my undergraduate studies and have stayed there since, co-authoring papers in digital health and computational materials science while building and shipping production machine learning systems. I care most about problems that combine rigorous methodology with real deployment, and I finish an accelerated Computer Science degree at the University at Buffalo in 2026 with a 3.8 GPA before starting my doctoral work.

Education

PhD in Computer Science

Starting Fall 2026

Michigan State University

East Lansing, MI

- Department of Computer Science and Engineering, Data Mining Laboratory. Advisor: [Dr. Pang-Ning Tan](#)
- Graduate Research Assistant, funded under an NSF award supporting *Enhancing Adversarial Robustness of Geospatio-Temporal Models*
- Research focus: spatiotemporal machine learning, deep learning based weather forecasting (DLWF), and adversarial robustness

Bachelor of Science in Computer Science, Minor in Statistics

August 2023 – June 2026

University at Buffalo, SUNY

Buffalo, NY

- Specialization in artificial intelligence. Coursework in reinforcement learning, computer vision, machine learning, quantum computing, and distributed systems
- GPA 3.8. Dean's List every semester. Completed the degree in three years on an accelerated course load (21 to 22 credits per semester)
- Presidential Scholarship, \$15,000 per year. PEARL Award, \$2,500 competitive undergraduate research grant, Experiential Learning Network

Research Experience

Graduate Research Assistant

Starting August 2026

Data Mining Laboratory, Michigan State University

East Lansing, MI

Adversarial robustness of geospatio-temporal deep learning models. Advisor: [Dr. Pang-Ning Tan](#).

- Joining an NSF-funded research group studying adversarial vulnerabilities in deep learning based weather forecasting (DLWF) models and developing defense methods for geospatio-temporal (GST) systems.
- Planned work builds on the lab's wavelet decomposition based localized targeted attack methods and downstream trajectory manipulation attacks, extending toward detection and defense frameworks for GST models.

Research Assistant

June 2024 – August 2026

Embedded Sensing and Computing (ESC) Group, University at Buffalo

Buffalo, NY

Mobile sensing for healthcare and rehabilitation. Advisor: [Prof. Wenyao Xu](#).

OralScan: AI-Assisted Geriatric Oral Screening

June 2024 – February 2025

- Built a full-stack mobile health platform in React Native and React, integrating a **YOLOv8** vision pipeline for guided intraoral image capture, multi-class dental disease screening, and automated tooth numbering across repeated scan sessions.
- Designed a spatiotemporal scan-guidance method that aggregates sequential intraoral frames to produce consistent disease detection signals despite user-induced motion.

- Co-authored a formative usability and acceptability study, submitted to *Smart Health* and currently under review. Presented findings to clinical stakeholders at CTSI and SURC 2024.
- Placed second (\$1,000) at the University at Buffalo Health Futures Challenge, Spring 2026.

OrthoScan: Orthodontic Remote Monitoring

February 2025 – October 2025

- Built mobile and backend systems for at-home orthodontic tracking, using **YOLO**-based hardware detection combined with depth-assisted spatial measurement to quantify structural change across longitudinal visits.
- Reconstructed 3D dental structures from intraoral scans using pretrained **GNNs** for mesh processing, and computed tooth position changes in both 2D image space and 3D world coordinates.
- Converted sequential intraoral imagery into calibrated geometric progress signals, enabling quantitative remote monitoring between office visits and reducing dependence on subjective clinician assessment.

mRehab: Telerehabilitation Platform (mrehab.agency)

November 2025 – August 2026

- Set up and configured **AWS** backend infrastructure for testing and staging environments, supporting AI agent capabilities within the live application.
- Built a real-time sensor logging service that ingests and monitors wearable sensor streams during active therapy sessions.
- Implemented **Kalman filters** for sensor fusion and activity detection, improving signal quality and exercise recognition under noisy real-world IMU data.
- Contributed to session logging, multi-modal sensor ingest pipelines, and exercise scoring algorithms currently running in live outpatient therapy workflows.

Undergraduate Researcher

June 2025 – July 2026

Peng Research Lab, University at Buffalo

Buffalo, NY

Computational materials science and clean energy applications. Advisor: [Prof. Jiayu Peng](#).

Perovskite Ordering and Symmetry-Aware Graph Neural Networks

June 2025 – November 2025

- Benchmarked symmetry-aware **GNN** architectures, including **ALIGNN**, on ordering-dependent energetics in crystalline perovskites. Contributed benchmarking results and revised analysis to arXiv:2409.13851 and the open-source **PerovskiteOrderingGCNNs** repository.
- Migrated experiment tracking infrastructure from SigOpt to **Weights & Biases**, and automated training loops, hyperparameter sweeps, and multi-architecture evaluation, reducing manual overhead and improving reproducibility.
- Co-authored a commentary on agentic AI for multimetallic catalyst discovery, published on *ChemRxiv* in 2025.

Equivariant MLIPs and Alchemical Extensions

December 2025 – July 2026

- Extended a dual-topology alchemical graph construction, originally built for **MACE**, to two additional equivariant **MLIP** architectures, fairchem **eSEN** and Meta **UMA**, enabling differentiable composition interpolation over disordered structures without retraining.
- Designed a lambda-weighted message-passing and expert-routing scheme for UMA’s **Mixture-of-Linear-Experts** backbone, correcting double-counting in composition-dependent expert gating and adding an endpoint-collapse mechanism for exact pure-composition limits.
- Found and fixed a reference-energy bug that caused non-physical energy discontinuities of roughly 15 eV near pure compositions. The bug was invisible to force-based relaxation but critical for free-energy calculations. Added regression tests to prevent recurrence.
- Built an automated verification suite of more than 12 tests, checking energy, force, stress, and gradient agreement to about 10^{-7} eV, along with benchmark figures and formula-level documentation cross-validating **MACE**, **eSEN**, and **UMA**.

Publications and Preprints

- **Dandapat, K.**, Liu, E., Master, T. A., Das, A., Bo, W., Cavuoto, L., and Xu, W. (2026). From Community Feedback to Measurement Redesign: Iterating mRehab for Accessible Home-Based Stroke Rehabilitation.

HumanSys 2026. Preprint.

- **Dandapat, K.**, Master, T. A., Das, A., Bo, W., Lei, M., Cavuoto, L., Subryan, H., and Xu, W. (2026). Separating Safety from Preference: A Role-Gated Evaluation Platform for Clinical Rehabilitation LLMs. *HumanSys 2026*. Preprint.
- Lei, M., Das, A., Master, T. A., **Dandapat, K.**, Reddipogu, P., Xian, J., Rowe, V., Craft, L., Tabb, K., Cavuoto, L., Bhattacharjya, S., Jo, H. J., Subryan, H., Bo, W., and Xu, W. (2026). Towards AI Agents for Intelligent Voice-Driven Interaction in a Home-Based Stroke Rehabilitation System. University at Buffalo and Georgia State University. Preprint.
- **Dandapat, K.**, et al. (2026). Demo: mRehab, A Tangible, Offline-First Mobile Platform for Post-Stroke Upper-Limb Rehabilitation with an Edge-Cloud Voice Assistant. *MobiComm 2026* (Demo Track). Preprint.
- Soni, P., **Dandapat, K.**, Gherardi, A., Bo, W., Li, R., and Xu, W. (2025). OralScan, an AI-Powered Mobile Tool for Geriatric Oral Healthcare: A Formative Usability and Acceptability Study. Submitted to *Smart Health*. Under review.
- Peng, J., Liu, C., Luo, Y., and **Dandapat, K.** (2025). Accelerating Multimetallic Catalyst Discovery with Robotics and Agentic AI. *ChemRxiv*, version 1. DOI: [10.26434/chemrxiv-2025-13n3f](https://doi.org/10.26434/chemrxiv-2025-13n3f).
- Peng, J., et al. (2024). Learning Ordering in Crystalline Materials with Symmetry-Aware Graph Neural Networks. *arXiv* preprint [arXiv:2409.13851](https://arxiv.org/abs/2409.13851). Contributed benchmarking and revisions.

Research Presentations

- Oral presentation and live demo, CTSI Research Group of Doctors, Nurses, and Medical Students (April 2024)
- Oral presentation, SUNY Undergraduate Research Conference (SURC) (April 2024)

Selected Projects

PaddockPsychRL [[GitHub](#)]

Multi-Agent Reinforcement Learning

- Built a multi-agent reinforcement learning framework that integrates Formula 1 temporal performance signals into agent psychology models within **PettingZoo** cooperative environments, mapping real-world behavioral dynamics to learned policy structure.
- Constructed driver-specific agent profiles from **FastF1** 2025 lap-time consistency metrics, using temporal performance variability as per-agent action-noise scales to reflect decision-making under pressure.
- Benchmarked tabular Q-learning and **MAPPO** baselines against psychology-conditioned policies. Psychology-aware agents raised the Strategy Resilience Score from 0.54 to 0.63, a 16.7 percent improvement, under noisy multi-agent coordination.

Stack: Python, FastF1, PettingZoo, Ray RLlib, MAPPO, Q-learning

Other Projects

Computer Vision

- **Marine Guardian**: geometric classifier combined with **MobileNetV2** for ship detection in satellite imagery. Achieved 98.72 percent classification accuracy, benchmarked against EfficientNet and ResNet-50 baselines.
- **Density-Based Crowd Counting**: **CSRNet** trained in **PyTorch** for per-pixel density estimation in heavily occluded surveillance scenes, avoiding the failure modes common to bounding-box detectors.
- **Facial Expression Benchmark**: controlled comparison of CNN, ResNet-34, and Vision Transformer backbones for emotion recognition. Best configuration reached 87.5 percent top-1 accuracy.

Professional Experience

Teaching Assistant, CSE 341: Computer Architecture

January 2026 – May 2026

Department of Computer Science and Engineering, University at Buffalo

Buffalo, NY

- Ran weekly tutorial sessions on **Assembly** and **VHDL** for more than 30 students, live-coding solutions to problems in cache design, memory hierarchies, and instruction pipelining.
- Led recitations covering processor pipeline stages, hazard detection, and cache replacement policies. Devel-

oped worked examples and problem sets to reinforce lecture material.

- Held weekly office hours, graded exams and assignments, and supported more than 100 students on Piazza throughout the semester.

Tutor and Peer-Assisted Leader (PAL)

August 2024 – December 2025

Tutoring and Academic Support Services (TASS), University at Buffalo

Buffalo, NY

- Served as PAL for STA 119 (Introduction to Statistics). Designed weekly worksheets aligned to recent lectures and led two 50-minute collaborative sessions per week. Course feedback indicated roughly a 30 percent improvement in quiz performance.
- Provided one-on-one tutoring across seven courses, STA 119, CSE 115, CSE 116, CSE 220, MTH 141, and MTH 142, covering introductory programming, data structures, and calculus.

Honors and Awards

- **PEARL Award:** \$2,500 competitive undergraduate research grant, University at Buffalo Experiential Learning Network (November 2025)
- **Health Futures Challenge:** Second place (\$1,000) for OralScan, University at Buffalo (Spring 2026)
- **Cybersecurity Excellence:** Top 100 globally, Northeastern Cybersecurity C2C Finals. 10th place in world finals (2025)
- **Collegiate Lockdown:** One of the top two teams representing UB. 4th place in finals (2025)
- **Dean's List:** University at Buffalo (Fall 2023, Spring 2024, Fall 2024)
- **Presidential Scholarship:** \$15,000 per year

Competitions and Challenges

- IAA AppXcelerate Application, University at Buffalo (January to March 2025), OralScan
- Russell L. Agrusa CSE Student Innovation Competition (November 2024), team member, OralScan
- Aging Innovations Challenge (November 2024), team member, OralScan
- Community Champions for Disability Health Challenge (October 2024), OralScan

Technical Skills

- **Languages:** Python, JavaScript, Java, C++, Rust
- **Machine learning and AI:** PyTorch, TensorFlow, computer vision, deep learning, graph neural networks (GNNs), YOLOv8, Vision Transformer (ViT), reinforcement learning (MAPPO, Q-learning), Ray RLlib
- **Scientific ML and materials:** ALIGNN, machine learning interatomic potentials (MLIPs), MACE, E(3)-equivariant MPNNs, Weights & Biases, PyTorch Geometric
- **Software engineering:** React, React Native, Node.js, Django, Git, CUDA, OpenCV, NumPy, Matplotlib
- **Databases and cloud:** MongoDB, SQL, Firebase, SQLite, AWS, Google Colab, Firebase Hosting

Research Interests

- Spatiotemporal machine learning and deep learning based weather forecasting (DLWF)
- Adversarial robustness of deep learning weather forecasting models, including wavelet-based attack and defense methods for geospatio-temporal data
- Geometric and equivariant deep learning, and graph neural networks for structured data
- Computer vision and sequence modeling for healthcare and scientific applications
- Machine learning for materials science and scientific discovery